CHAPTER 1

# Simple Interest & Linear Accumulation

Axiomatic Foundations, Differential Formulations, and Step-by-Step Analytical Solutions

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## 1.1 Definition & Mathematical Axioms

Simple Interest represents the foundational linear valuation model in financial mathematics. The core axiom defining this accumulation regime is that interest accrued over any sub-interval is **never capitalized** into the earning principal; that is, interest does not generate further interest.

### Definition 1.1 (Simple Accumulation Function)

Let initial capital principal be $P > 0$, nominal annual interest rate be $r \geq 0$ (expressed as a decimal fraction: $r = R / 100$ for percentage rate $R\%$), and time horizon be $t \geq 0$ measured in years.
The **total accrued interest** $I(t)$ and **terminal accumulated value** $A(t)$ are defined by:

$$I(t) = P \cdot r \cdot t$$

$$A(t) = P + I(t) = P(1 + r \cdot t)$$

*The accumulation factor is denoted as $a(t) = 1 + r \cdot t$, which satisfies $a(0) = 1$.*

## 1.2 First-Principles Derivation & Differential Viewpoint

**Axiomatic Derivation:** Under simple accumulation, interest $I$ satisfies three simultaneous physical properties: (1) $I \propto P$ (principal scale invariance), (2) $I \propto r$ (yield homogeneity), and (3) $I \propto t$ (temporal linearity). Combining these yields $I = k \cdot P \cdot r \cdot t$, with proportionality constant $k = 1$ by convention.

**Ordinary Differential Equation (ODE) Perspective:**
Examining instantaneous wealth rate of change $dA/dt$ contrasts simple and continuous regimes:

| Valuation Regime | Governing Differential Equation | Closed-Form Solution $A(t)$ | Growth Dynamic |
|---|---|---|---|
| **Simple Interest** | $dA/dt = P \cdot r$ | $A(t) = P(1 + rt)$ | Constant slope (Linear) |
| **Continuous Compounding** | $dA/dt = r \cdot A(t)$ | $A(t) = P \cdot e^{rt}$ | Wealth-proportional (Exponential) |

### Theorem 1.1 (Sub-Period Dominance via Bernoulli's Inequality)

For any strictly positive rate $r > 0$ and fractional duration $t \in (0, 1)$, simple interest accumulation is *strictly greater* than annual compound interest accumulation:

$$\mathbf{1 + r \cdot t > (1 + r)^t \quad \forall\, t \in (0, 1)}$$

**Proof:** Let $f(t) = (1 + r)^t$ for $t \geq 0$. The second derivative is $f''(t) = [\ln(1 + r)]^2(1 + r)^t$. Since $[\ln(1+r)]^2 > 0$ and $(1+r)^t > 0$ for all $r > 0$, $f''(t) > 0$ strictly, confirming $f$ is strictly convex.

The secant line connecting $(0, 1)$ and $(1, 1+r)$ is $g(t) = 1 + rt$. By strict convexity, the graph of $f$ lies strictly below chord $g$ on the open interval $(0, 1)$. Thus, $(1 + r)^t < 1 + rt$. ■

## 1.3 Step-by-Step Worked Examples

Below are two formal case studies solved by hand: a multi-year sovereign bond yield and a money-market bridge facility with exact day counts.

### Example 1.1: Multi-Year Sovereign Treasury Bond

**Problem:** An investor commits principal capital of ***P* = INR 100,000** into a fixed-income treasury bond offering a simple interest rate of ***R* = 7.50% per annum**. The bond matures in ***t* = 3 years**. Calculate total interest accrued and final maturity amount.

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**Step-by-Step Analytical Solution:**

- **Step 1 (Parameter Normalization):** $P = 100{,}000$, $r = 7.50 / 100 = 0.075$, $t = 3.0$ years.
- **Step 2 (Interest Calculation):**
  $I = P \cdot r \cdot t = 100{,}000 \times 0.075 \times 3 = 100{,}000 \times 0.225 =$ **INR 22,500.00**
- **Step 3 (Terminal Wealth Calculation):**
  $A = P + I = 100{,}000 + 22{,}500 =$ **INR 122,500.00**
- **Step 4 (Annual Verification):**
  Year 1: $I_1$ = INR 7,500.00 | Year 2: $I_2$ = INR 7,500.00 | Year 3: $I_3$ = INR 7,500.00
  Total sum: 3 × 7,500 = INR 22,500.00 (verified exact linear accumulation).

### Example 1.2: Fractional Tenors & Day-Count Mechanics (The Banker's Rule)

**Problem:** A short-term commercial credit facility of ***P* = INR 250,000** is issued on **March 15** at an annual simple rate of ***R* = 9.20%**, and repaid on **October 28** of the same non-leap year. Compare interest accrued under:

- **Method A:** Exact / Actual/365 Convention (Common in sovereign bond markets)
- **Method B:** Banker's Rule / Actual/360 Convention (Standard in US money markets and commercial paper)

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**Step-by-Step Analytical Solution:**

- **Step 1 (Exact Elapsed Days):**
  March (31 – 15) = 16 | Apr = 30 | May = 31 | Jun = 30 | Jul = 31 | Aug = 31 | Sep = 30 | Oct = 28
  **Total days elapsed:** $d = 16 + 30 + 31 + 30 + 31 + 31 + 30 + 28 =$ **227 days**.
- **Step 2 (Method A: Actual/365 Convention):**
  Annualized duration fraction: $t_{365} = 227 / 365 \approx 0.6219178$ years
  $I_{365} = 250{,}000 \times 0.092 \times (227 / 365) = 23{,}000 \times 0.6219178 =$ **INR 14,304.11**
  Terminal repayment: $A_{365} = 250{,}000 + 14{,}304.11 =$ **INR 264,304.11**.
- **Step 3 (Method B: Actual/360 Convention — The Banker's Rule):**
  Annualized duration fraction: $t_{360} = 227 / 360 \approx 0.6305556$ years
  $I_{360} = 250{,}000 \times 0.092 \times (227 / 360) = 23{,}000 \times 0.6305556 =$ **INR 14,502.78**
  Terminal repayment: $A_{360} = 250{,}000 + 14{,}502.78 =$ **INR 264,502.78**.
- **Step 4 (Yield Discrepancy & Economic Analysis):**
  Absolute difference: $\Delta I = I_{360} - I_{365} = 14{,}502.78 - 14{,}304.11 =$ **INR 198.67**
  Ratio: $I_{360} / I_{365} = 365 / 360 \approx 1.013889$ (+1.39%).
  *Takeaway:* The Actual/360 convention artificially magnifies the lender's effective annual yield by exactly 1.39%, demonstrating why institutional term sheets explicitly define day-count conventions.

## 1.4 Algorithmic Implementation in Python

Below is the complete implementation from models/time_value.py. It provides domain validation, typing, and clear docstrings for modular inclusion in the decision-support system:

**Listing: models/time_value.py — Simple Interest Engine**

```
def simple_interest(principal: float, annual_rate: float, time_years: float) -> float:
    """
    Computes accrued simple interest: I = P * r * t.
    Parameters:
        principal (float): Initial capital (P >= 0).
        annual_rate (float): Nominal annual rate as decimal (r >= 0).
        time_years (float): Duration in years (t >= 0).
    """
    if principal < 0.0 or annual_rate < 0.0 or time_years < 0.0:
        raise ValueError("Financial parameters must be non-negative.")
    return float(principal * annual_rate * time_years)


def simple_interest_amount(principal: float, annual_rate: float, time_years: float) -> float:
    """Computes terminal accumulated wealth: A = P * (1 + r * t)."""
    return float(principal + simple_interest(principal, annual_rate, time_years))
```

## 1.5 Mathematical Properties Summary

| Mathematical Property | Specification & Value | Analytical Significance |
|---|---|---|
| **Domain & Constraints** | $P > 0$, $r \geq 0$, $t \geq 0$ | Strict non-negativity across economic state variables |
| **Accumulation Function** | $a(t) = 1 + rt$ | Affine linear operator with slope equal to nominal rate $r$ |
| **Marginal Accrual Rate** | $da/dt = r$ | Invariant to elapsed time $t$ and current accumulated wealth |
| **Convexity / Curvature** | $d^2a/dt^2 = 0$ | Zero second derivative; absence of compounding curvature |